

PREFACE

The Sindh Textbook Board, is assigned with preparation and publication of the textbooks to equip our new generation with knowledge, skills and ability to face the challenges of new millennium in the fields of Science, Technology and Humanities. The textbooks are also aimed at inculcating the ingredients of universal brotherhood and to reflect the valiant deeds of our forebears and portray the illuminating patterns of our rich cultural heritage and traditions.

To accomplish this noble task, a team of educationists, experts, working teachers and committee of friends has been constituted which incessantly endeavours to develop, test and improve content and design of the textbooks on the basis of the feed back. The present intellectual product has been presented with joint efforts of the team to project the aforementioned assignment in its true perspective.

It is expected that the discerning parents, learned teachers and concerned citizens will go through it and offer their valuable opinion to the Board for bringing about improvement in the next edition.

Chairman
SINDH TEXTBOOK BOARD

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

Chapter

1

The Universe

Allah Almighty has created this Earth, the Sun, the Moon, and the planets for the benefit of mankind. His creation makes our belief in His existence still stronger.

This Universe is so large that it is almost impossible for the human mind to perceive it in its entirety. Everyone understands the Universe and its effects on us in a different way. But most of us would agree that humans have an innate need to understand their surroundings. While philosophers delve into the purpose of the Universe, scientists investigate its mechanism, pointing telescopes at massive objects which are far away at unimaginable distances or observing the behaviour of tiny drops of liquid in the laboratory. They strive to understand how the Universe is put together and how does it work. However, for our need and understanding we will be mainly looking at the Sun, the moon and the planets which together constitute the 'Solar System'. It is also interesting to know that our Solar System is one of the many such Systems in the Universe.

Solar System

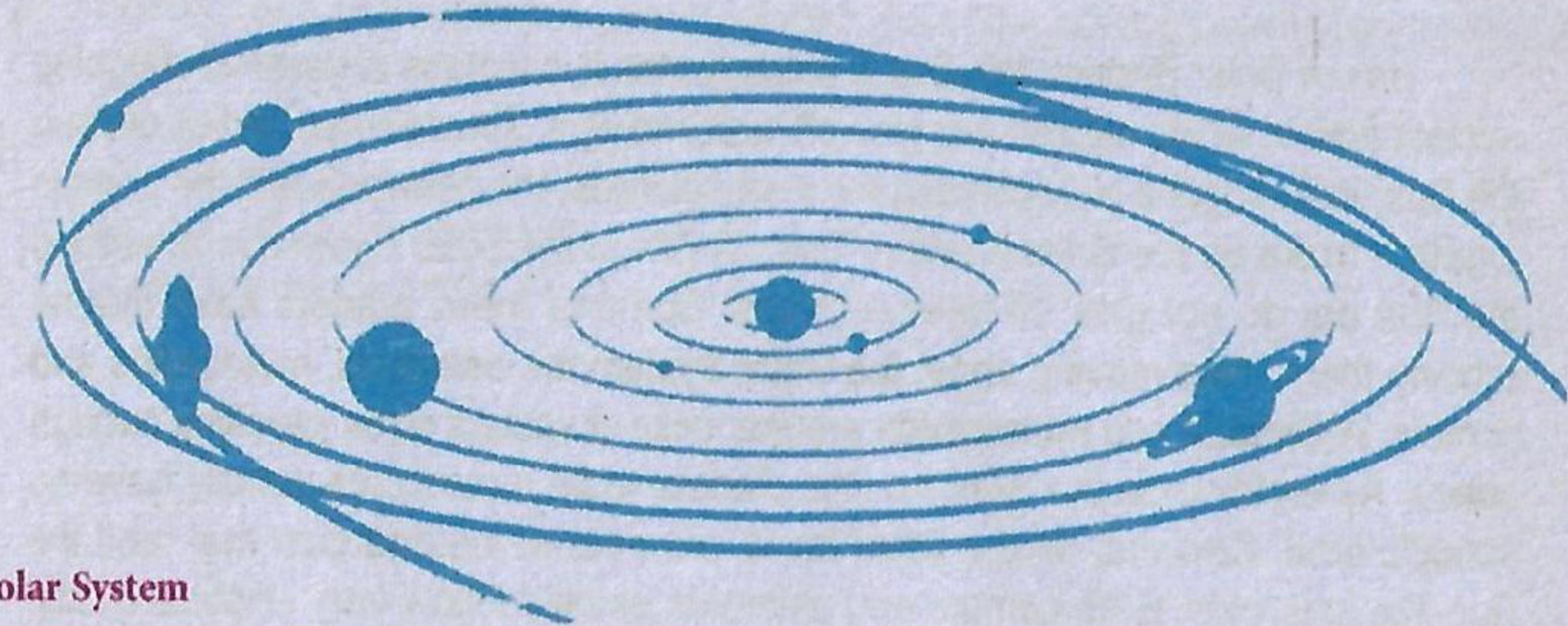
In our Solar System, the Sun is the only star. It is a mass of gases undergoing nuclear fusion, which causes it to give off heat and light. The celestial bodies orbiting the Sun, including the eight planets and their satellites, the asteroids and the comets together make up the Solar System. The planets of the Solar System orbit around the Sun but do not give off heat and light. Some of these planets have moons orbiting them. Also moving about the Solar System are asteroids, meteoroids and comets. **Asteroids** and **meteoroids** are like rocks of various sizes travelling through space. Asteroids orbit the Sun like little planets while meteoroids usually have no specific orbit. **Comets**, on the other hand, are balls of ice and dust that orbit the Sun like asteroids. Both comets and asteroids usually follow very elliptical orbits.

Eight major planets are currently known. They are commonly divided into two groups: the inner planets (Mercury, Venus, Earth, and Mars) and the outer planets (Jupiter, Saturn, Uranus, and Neptune). The inner planets are small and are composed primarily of rock and iron. The outer planets are much larger and

consist mainly of hydrogen and helium. Pluto does not belong to either group, and there is an ongoing debate as to whether Pluto should be counted as a major planet or not.

Mercury is surprisingly dense, apparently because it has an unusually large iron core. With only a transient atmosphere, Mercury has a surface that still bears the record of bombardment by asteroidal bodies early in its history. Venus has a carbon dioxide atmosphere 90 times thicker than that of the Earth, causing an efficient greenhouse effect. The surface temperature is the hottest of any planet — about 477°C. Earth is the only planet with abundant liquid water and known life. Strong evidence exists that Mars once had liquid water on its surface, but now its carbon dioxide atmosphere is so thin that the planet is dry and cold, with polar caps of frozen water and solid carbon dioxide, or dry ice.

Jupiter is the largest of the planets. Its hydrogen and helium atmosphere contains pastel-coloured clouds, and its immense magnetosphere, rings, and satellites make a Planetary System of its own. One of Jupiter's largest moons, Io, has volcanoes that produce the hottest surface temperatures in the Solar System. At least four of Jupiter's moons have atmospheres, and at least three show evidence that they contain liquid or partially-frozen water. Saturn rivals Jupiter, with a much more intricate ring structure and a similar number of satellites. One of Saturn's moons,



Solar System

Titan, has an atmosphere thicker than that of any other satellite in the Solar System.

Uranus and Neptune are deficient in hydrogen compared with Jupiter and Saturn; Uranus, also ringed, has the distinction of rotating at 98° to the plane of its

orbit. Pluto seems similar to the larger, icy satellites of Jupiter or Saturn. Pluto is so distant from the Sun and so cold that methane freezes on its surface.

Some important data about the planets of the Solar System

Serial Number	Name of the Planet	Diameter in Kilometres	Distance from Sun (in Kilometres)	Period of revolution round the Sun.	General Observation
1	Mercury	4880	5,79,00,000	88 days	Smallest and nearest to Sun
2	Venus	12104	10,82,00,000	225 days	Nearest to the Earth
3	Earth	12740	14,96,00,000	365 ¼ days	
4	Mars	6787	2,27,00,000	687 days	
5	Jupiter	137600	74,83,00,000	11 ¾ years	Biggest Planet
6	Saturn	12000	1,42,58,00,000	29 ½ years	
7	Uranus	51820	2,86,94,00,000	84 years	
8	Neptune	49500	4,49,16,00,000	164 ¾ years	

All these planets, which are millions of kilometres away from the Sun and are revolving around it have their own gravity or gravitational force. Due to this gravitational force these planets maintain distance from the Sun and from each other. Each planet moves around the Sun on its fixed orbit. The force that keeps

all of them in their own places and does not allow them to move away from their regular path, is the gravity of the planets through which they attract or pull other objects towards them.

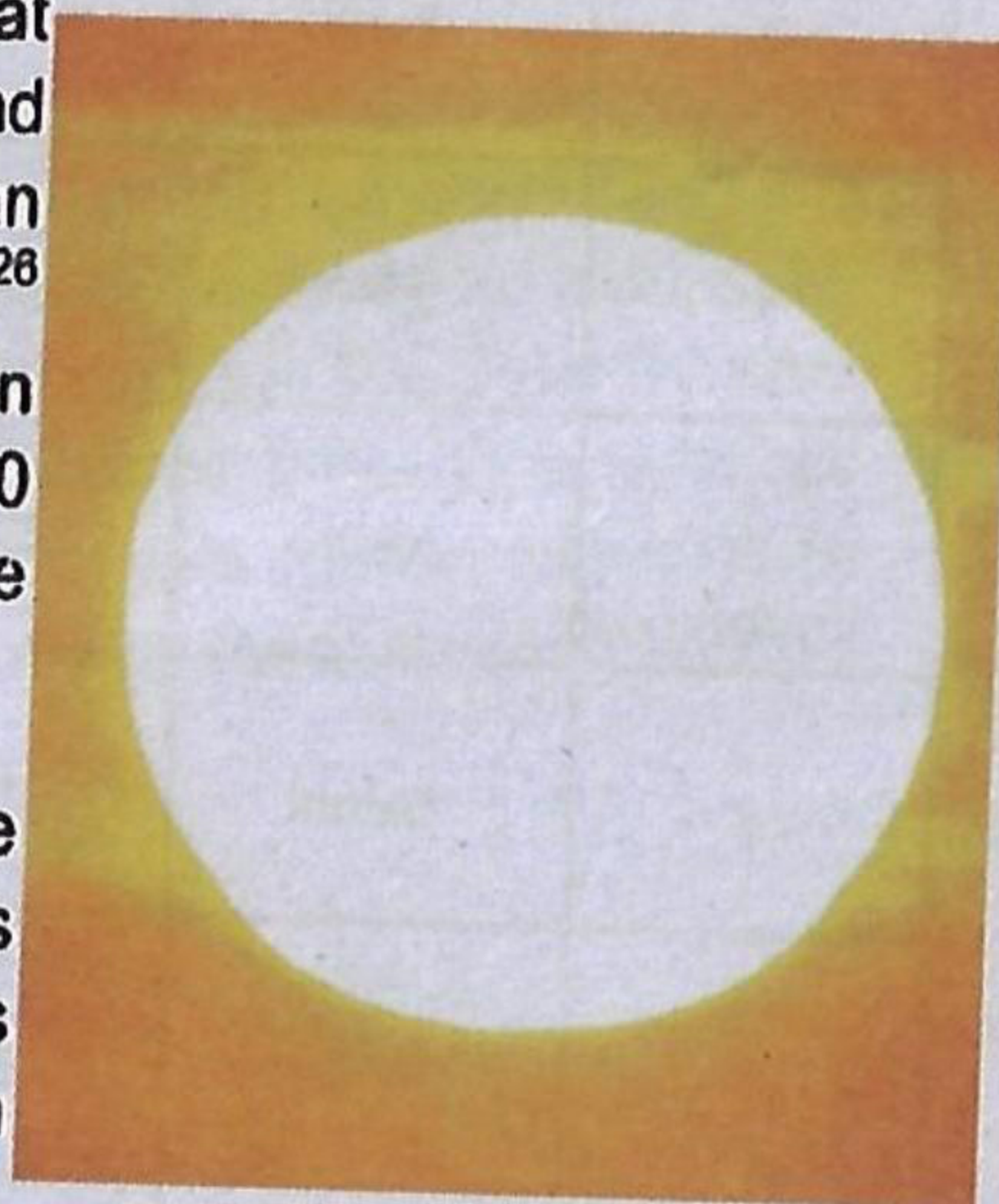
The Sun

The Solar System is named after the Sun, which is the centre of this System. The Sun, which is a star has its own light. The Sun is from the family of 'galaxy'. Galaxies are made up of nebulae, billions of stars, and all of the objects that are orbiting those stars.

There are three main types of galaxies: **irregular, spiral, and elliptical.**

Sun is the closest star to the Earth. It is a huge mass of hot, glowing gas. The strong gravitational pull of the Sun holds Earth and the other planets in the Solar System in orbit. The Sun's light and heat influence all of the objects in the Solar System and allow life to exist on Earth. The Sun produces an enormous amount of light. It generates 3.83×10^{26} watts of power in the form of light and heat. In comparison, an incandescent lamp emits 60 to 100 watts of power. The temperature of the outer, visible part of the Sun is 5500°C .

From the Earth the Sun looks small, because it is far away. Its average distance from Earth is 149.6 million kilometres. Light from the Sun takes about eight minutes and twenty seconds to reach the Earth.



The Sun

The Sun's gravitational pull holds the Solar System together. The planets, asteroids, comets, and dust that make up our Solar System are strongly attracted to the Sun's huge mass. This gravitational attraction keeps these bodies in orbit around the Sun.

Large, dark spots, called Sunspots, are often visible in the photosphere. The photosphere is the lowest, densest level of the Solar atmosphere. The visible light that reaches the Earth from the Sun originates in the photosphere. The biggest Sunspots exceed Earth in size and are easily visible with a telescope. Sunspots rotate with the Sun and change in size and shape. They come and go, with lifetimes

lasting from hours to months. The number of Sunspots increases, then decreases, over an 11-year cycle. A Sunspot is about ten times brighter than the full Moon.

The Sun provides the Earth with vast amounts of energy every day. The oceans and seas store this energy and help keep the temperature of the Earth at a level that allows a wide variety of life to exist. Plants use the Sun's energy to make food, and provide food for other organisms. The Sun's energy also creates wind in Earth's atmosphere. This wind can be harnessed to produce power. While it lights our day and provides energy for life, sunlight can also be harmful to people. Human skin is sensitive to ultraviolet light emitted from the Sun.



The spots on the sun

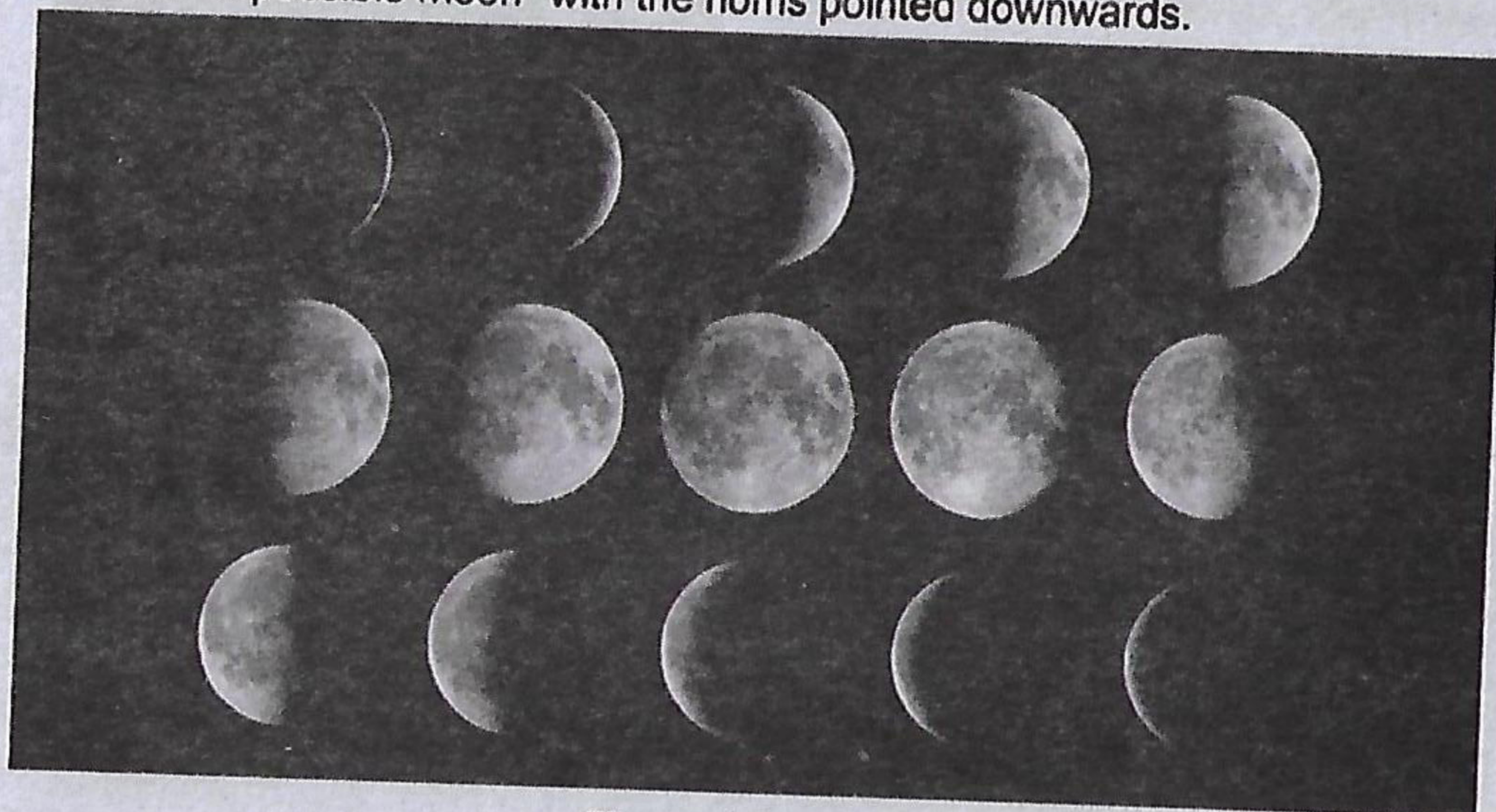
The Moon

Moon is the most important of the heavenly bodies in the Solar System. Man has travelled on moon and has not only placed its feet on it but also walked on its surface. The US spacecraft Apollo 11 was launched on July 16, 1969. Neil Armstrong, Michael Collins and Edwin Addrin went on the first ever moon-landing flight. The lunar module touched Moon's surface on July 20, 1969. Armstrong and Addrin not only landed on the moon's surface but also installed UN flag and scientific devices and equipment and remained on the surface of the moon for 21 hours and 37 minutes. Michael Collins orbited the moon as he stayed on the spacecraft.

Moon, the Earth's only natural satellite is fifth in diameter among planetary satellites, more than two-thirds as large as Mercury, and more than three times the diameter of the largest asteroid. It is, in fact, over one-fourth the size of the Earth, with a diameter of 3475 kilometres. The Moon moves about the Earth at an average distance of 384,403 kilometres and at an average speed of 3,700 kilometres per hour. It completes one revolution in an elliptical orbit about the Earth in 27 days 7 hours 43 minutes. The phases of the moon are caused by the relative positions of the Earth, the Sun, and the Moon.

The Sun always illuminates half of the moon facing it. When the Sun and the moon are on opposite sides of the Earth, the moon appears "full" to us, a bright, round disk. When the moon is between the Earth and the Sun, it appears dark, a "new" moon. In between, the moon's illuminated surface appears to grow (wax) to full, then decreases (wanes) to the next new moon.

The edge of the shadow is always curved, being an oblique view of a circle, giving the moon its familiar crescent shape. Because the "horns" of the moon at the ends of the crescent are always facing away from the setting or rising Sun, they always point upward in the sky. It is fun to watch for paintings and pictures which show an "impossible moon" with the horns pointed downwards.

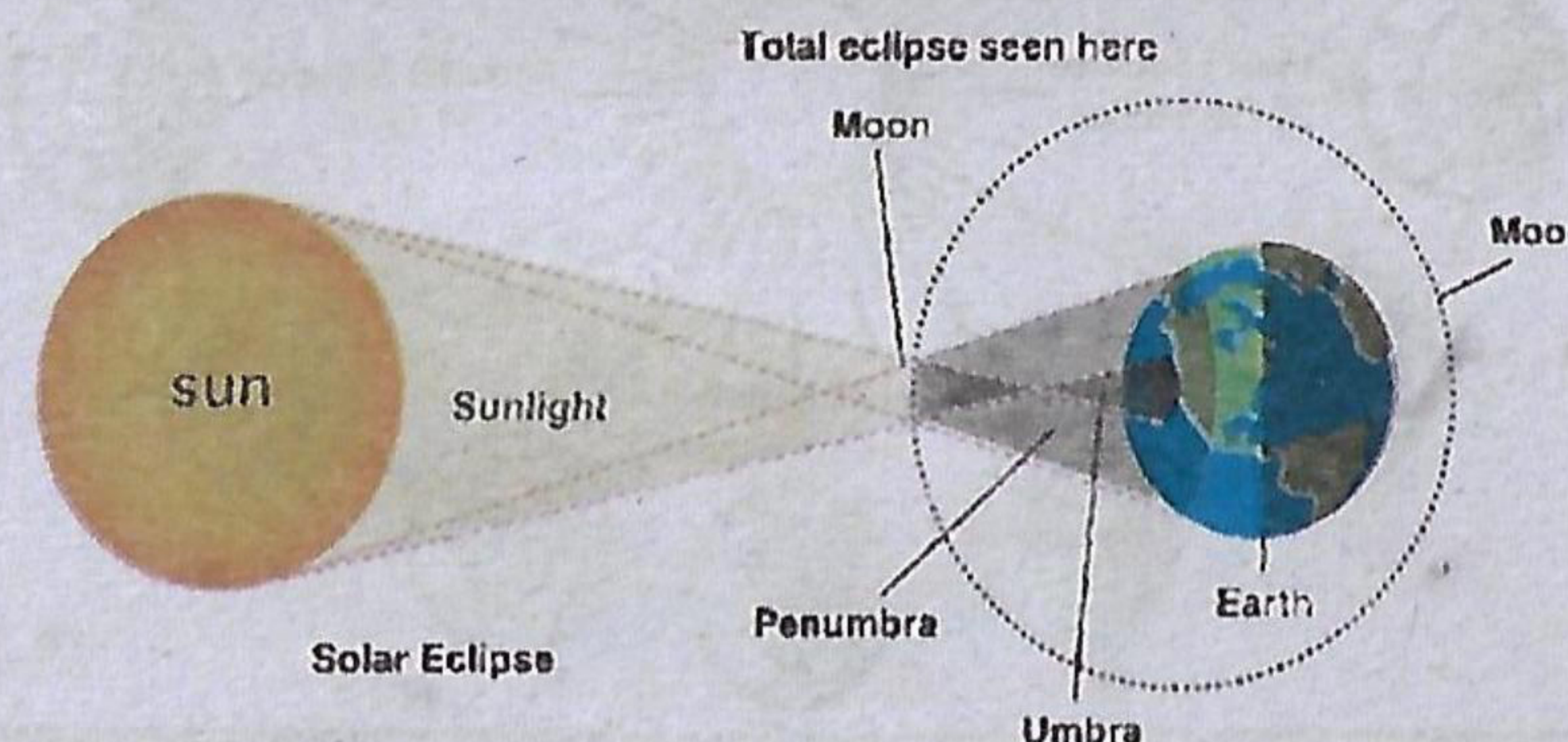
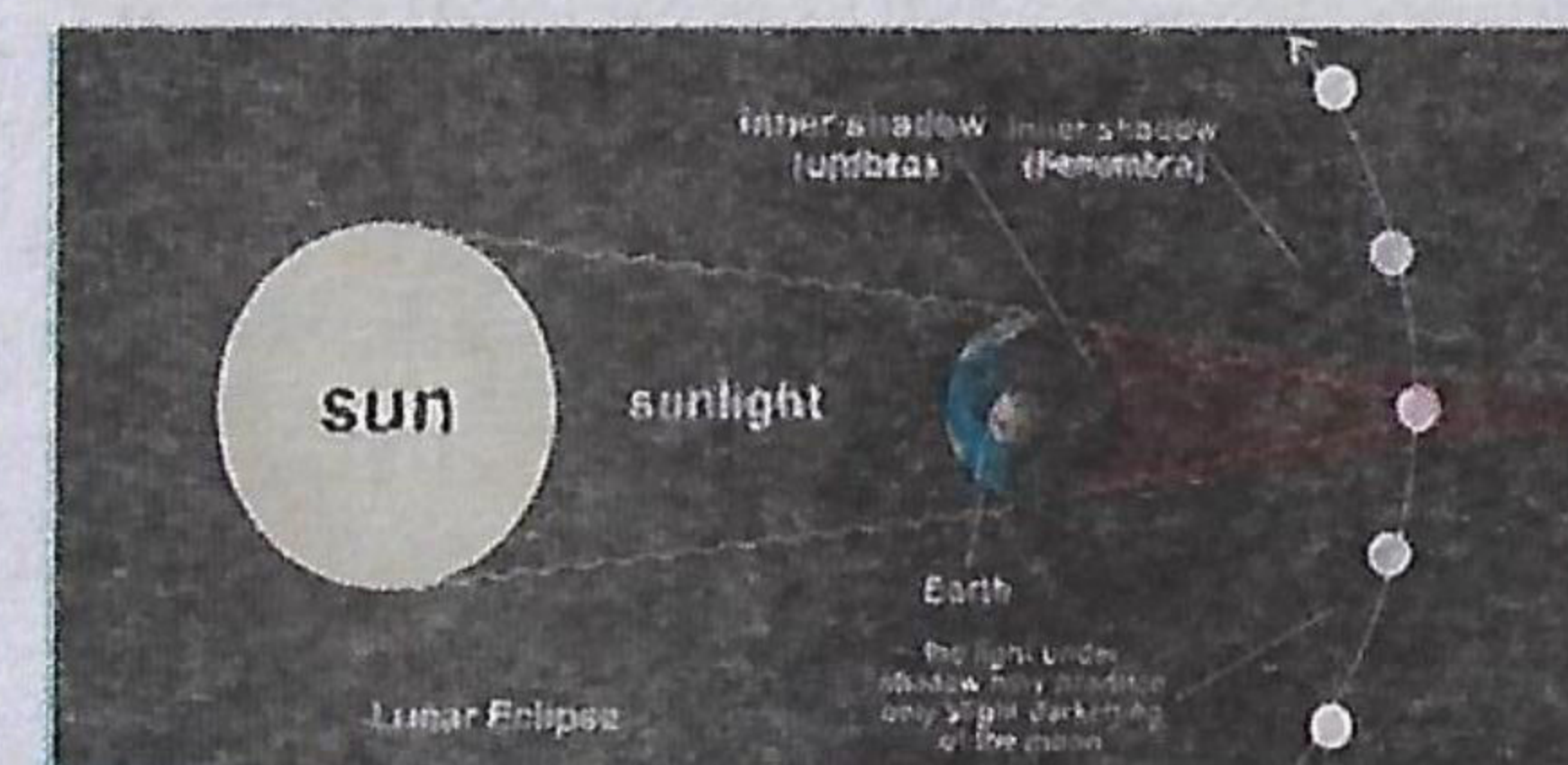


Phases of the Moon

People often refer to "the dark side of the moon", but there is no such thing. The Sun shines on all sides of it in turn. However, there is a "far side of the moon" which is never seen from the Earth.

Eclipse

Eclipse, in astronomy means the obscuring of one celestial body by another, particularly that of the Sun or a planetary satellite. Two kinds of eclipses involve the Earth: those of the moon, or lunar eclipses; and those of the Sun, or Solar eclipses. A Lunar eclipse occurs when the Earth is between the Sun and the moon and its shadow darkens the moon. A Solar eclipse occurs when the moon is between the Sun and the Earth and its shadow moves across the face of the Earth.

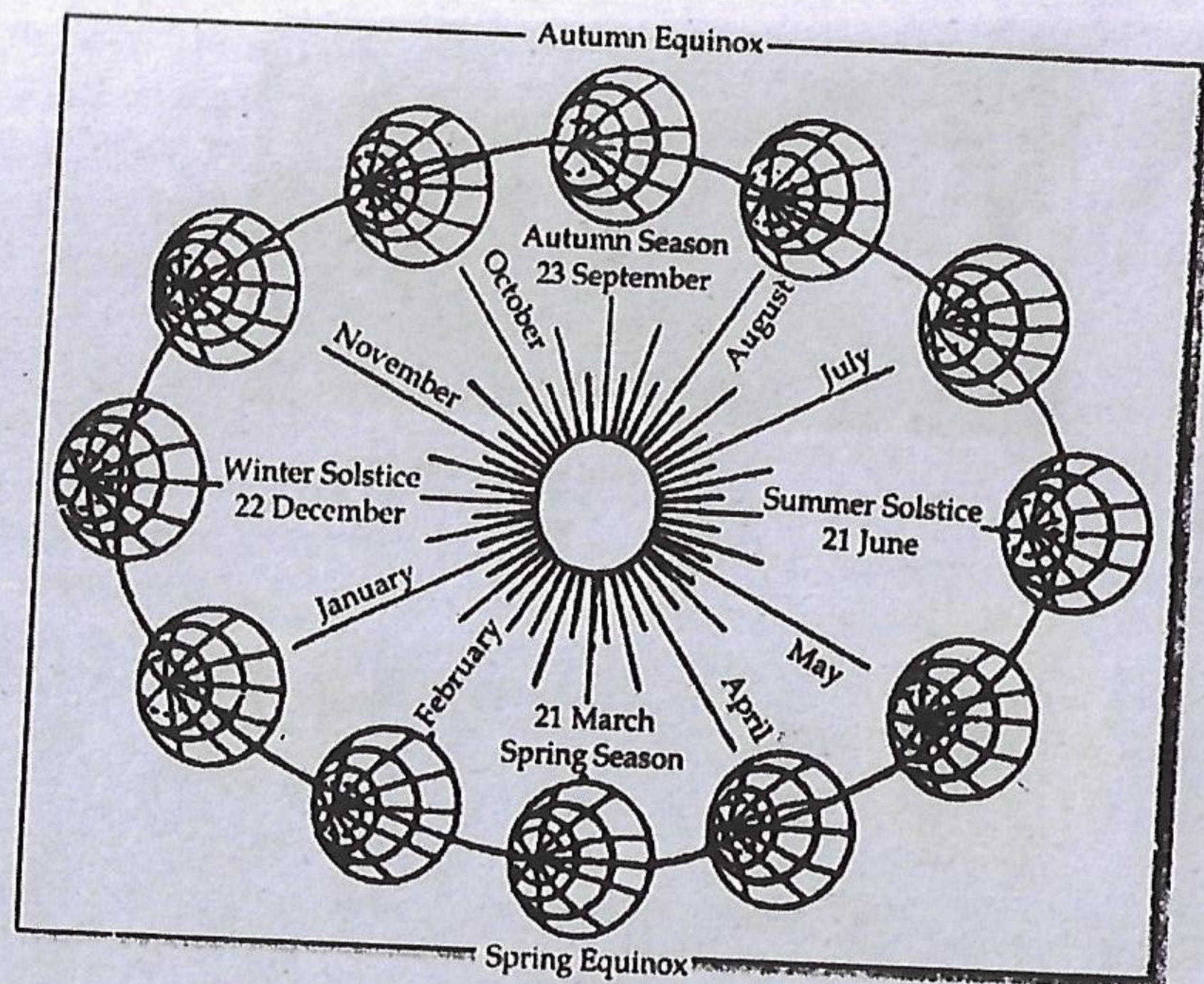


Place of the Earth in the Solar System

The Earth is one of the planets in the Solar System. It is the third in distance from the Sun and the fifth largest of the planets in diameter. The mean distance of the Earth from the Sun is 149.6 million kilometres. It is the only planet known to support life, although some of the other planets have atmosphere and contain water.

The Earth completes its revolution around the Sun in 365 $\frac{1}{4}$ days. The changes of weather on Earth are caused because of the changing position and distances of Earth during this revolution (around the Sun). A year is of 365 days, we leave the fraction $\frac{1}{4}$ and after every four years a complete day is added to the year and February is of 29 days, this is called a "Leap Year". When a Century year (say 1600 or 2000) occurs, a complete day (29th February) is added only if the Century year gets fully divided by 400. The year 2000 had 28 days in February. The next such event will take place in the year 2400. The Earth rotates on its axis once every 24 hours. Days and nights are caused due to this movement.

The outstanding feature of the Earth is that it has the presence of liquid water. Water is vital not only for the biosphere but also for the geologic processes of erosion, transport, and deposition that shape the Earth's surface. Yet, if the Earth were closer to the Sun, the water would have vapourized; if farther, it would have



The Axis of the Earth

turned into ice. Two-thirds of the terrestrial surface is covered by oceans. It was long believed that the continents, constituting the remaining one-third of the surface, had remained in a fixed position throughout the Earth's history. Gradually, some Earth scientists dared to suggest that there had been major continental displacements, and finally, during the 1960s, investigators developed the full picture of seafloor spreading and plate tectonics. The continents, though constantly in motion, are in fact the oldest portions of the Earth's surface, for the seafloor is created at ridges and consumed at trenches on a geologically short time scale. Other planets, notably Mars and Venus, have surface features that suggest some elements of plate tectonics.

Axis of the Earth

The path on which the Earth revolves is called 'orbit'. The orbit of the Earth is elliptical or egg-shaped. It is because of the shape of the orbit that the distance of the Earth from the Sun is not constant and also that the Earth rotates on its axis not in a vertical but slanting position. The Earth's axis is tilted at 66 1/2 to the plane of its orbit. Due to the elliptic shape of the orbit and tilt of the axis there are different seasons and the length of the days and the nights is not equal throughout the year.

We all experience four seasons in a year with changed weather conditions. At different times of the year, the duration of days and nights are different. We have longer days and shorter nights in summer and shorter days and longer nights in winter. Had the Earth's orbit been round, there would have been only one season and if its axis not been tilted but vertical, days and nights would have been of equal duration throughout the year.

The Earth's movement on its orbit, which causes seasons, is called 'revolution' and its movement on its axis which causes days and nights is called 'rotation'.

Exercise

A. Answer the following questions:

1. What is a galaxy?
2. How many Solar Systems are there in the Universe?
3. Find out the effects of eclipses on the Earth and on the human beings.
4. How are seasons caused?
5. What is a Lunar calendar?

Activities

1. Make a model of the Solar System as a group activity and display your model in the class. Discuss it with your class teacher and use different materials for making the model.
2. Conduct a research on the launching of spacecrafts on the surface of the moon.

3. Take a torch, a small ping-pong ball and a globe. Throw the light of the torch on the globe placing the ping-pong ball in the centre.
4. Observe the phases of moon for a month and discuss it in the Class.

Extension Activity

1. Social Studies teacher to arrange a field trip for students to the PIA Planetarium at Karachi. Ask students to write an essay focusing on their newly acquired learning.



Chapter

2

The Division of Land and Water on the Earth

The division of land and water is not equal on the Earth. The area covered with water is comparatively more than the area covered by land. 71 per cent of the Earth's surface is covered with water and the land is spread over the remaining 29 per cent. The large masses of water are called oceans and large masses of land are called continents.

The Northern Hemisphere mainly comprises land where as the Southern Hemisphere is mainly covered with water.

The land is divided into big regions, which are called continents. Each continent has its own features, which make it different from the other continents. On the map of the world the division of land and water, i.e. oceans and continents is shown. There are seven continents in the world. Their names are:

- | | | |
|------------------|------------------|--------------|
| 1. Asia | 2. Europe | 3. Africa |
| 4. North America | 5. South America | 6. Australia |
| 7. Antarctica | | |

The geographical and political importance of the continents is discussed below:

Asia

According to area and population Asia is the largest continent of the world. Asia is bounded on three sides by oceans: the Arctic to the north, the Pacific to the east, and the Indian to the south. Many seas, bays, and gulfs indent the continent's coastline, which is 62,000 kms. long. Ural mountain ranges and Caspian Sea separate Asia from Europe. Similarly Red Sea and Suez Canal separate Asia from Africa.

Asia's mightiest mountain ranges radiate in great sweeping arcs from the Pamirs of Central Asia, a highland region where Tajikistan, Afghanistan, and China intersect. In the Southeast of the Pamirs are the Himalayas, spanning 2,400 kms. from the border between India and Pakistan in the west, to the border between India and Myanmar in the east. The Karakoram Range lies just north of the western Himalayas. These two ranges contain two of the world's highest peaks, including Mount Everest, which lies on the border between Tibet and Nepal.